IMMERSIONS ARE MONOMORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4861-4870, following the statement at 4855-4859.

Editorial note. Both routes are spelled; 'mono is closed under composition' is the trivial category fact used silently in route two.

Allowed standard prerequisites (editorial).

- Injective on points with surjective stalks is mono. lemma-injective-points-surjective-stalks (card L26421, cited at ~4863). First route: check the criterion.
- Restricting maps to opens / factoring through closed subspaces. lemma-restrict-map-to-opens (same chapter, cited at ~4865) and lemma-characterize-closed-subspace (card L26388, cited at ~4866). Second route: open and closed immersions are mono.

Source reference. Original lemma 01HP: sources/stacks-schemes/pinned-original-source.tex, source lines 632–632. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01L6: sources/stacks-schemes/pinned-original-source.tex, source lines 4828–4828. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HI: sources/stacks-schemes/pinned-original-source.tex, source lines 516–516. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *An immersion of schemes is a monomorphism. In particular, any immersion is separated.*

Proof. We can see this by checking that the criterion of Lemma 01L6 applies. More elegantly perhaps, we can use that Lemmas 01HI and 01HP imply that open and closed immersions are monomorphisms and hence any immersion (which is a composition of such) is a monomorphism. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.